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An integrated approach to the elicitation of customer requirements for engineering design using picture sorts and fuzzy evaluation

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Abstract

Product definition is widely accepted as one of the key factors to be considered in the early stage of product design and development. It has a direct impact on the success of a new product in the marketplace. Frequently, product definition is solicited through the voice of the customer (VoC). As such, an organization will obtain a prominent competitive edge over its competitors if it is able to effectively capture the genuine VoC or the requirements of a customer. Sorting techniques provide a means to elicit customer requirements. Although a number of sorting techniques were developed, few of them are able to handle the uncertainty and imprecision inherited from the VoC and subjective judgment because of their crisp decision making. An integrated approach is described, which is based on repeated single-criterion sorts and fuzzy evaluation, for the elicitation of customer requirements. The details of the integrated approach are presented. The approach is validated using a case study on the design of a wooden golf-club head. The details of the validation are discussed.

Keywords: Customer Requirements Acquisition; Fuzzy Evaluation; Knowledge Acquisition; Picture Sorts; Repeated Single-Criterion Sorts; Sorting Techniques

1. INTRODUCTION

Customer requirements elicitation and evaluation are critical procedures and essential premises for successful new product development (NPD). During the early stage of product concept development, these are also prominent metrics for an organization to improve its competitive edge in the marketplace. Hence, an organization should put forth considerable effort in capturing the genuine or "real" voice of the customer (VoC) rather than technological issues during this stage of development. Because of the verbatim origin of customer requirements that may possess a tacit and nontechnical nature, customer requirements are difficult for elicitors, such as knowledge engineers, to handle. Accordingly, the so-called requirements acquisition bottleneck in capturing the VoC may be due to the following:

- the application of a single method to deal with a complex requirement acquisition problem;
- the unavailability or lack of expert guidance in eliciting and analyzing the VoC; and
- the application of a quantitative evaluation for qualitative items.

Quality function deployment (QFD) is a widely used technique for the integration of marketing and engineering functions during NPD. Inevitably, customer requirements elicitation is the first key stage of QFD that will determine the success level of product concept development. To enhance the effectiveness of QFD, Burchill and Fine (1997) applied the Kawakita Jiro (KJ) method (affinity diagram) as a contextual link between the initial and well-understood customer requirements. Fung et al. (1998) adopted a so-called analytic hierarchy process (AHP) to analyze and prioritize customer requirements prior to their being employed as customer attributes in a house-of-quality matrix. However, the subjective factors and the intensive statistical analysis involved during customer requirements acquisition may complicate the problem and even lead the elicitors astray.

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Many techniques, such as fuzzy systems (Turksen & Willson, 1992), regression analysis (Jenkins, 1995), and expert systems (Hauge & Stauffer, 1993), were investigated to elicit customer requirements more accurately and objectively. Among these efforts, Hauge and Stauffer (1993) developed the taxonomy of customer requirements as an initial concept graph structure for questionnaires used for an expert system. The approach was aimed at further elicitation of knowledge from customers. Chen et al. (2001*a*) established a concurrent product design evaluation system, which is an expert system for wood golf club head design. In developing this system, a graph decomposition algorithm (Chen & Occeña, 2000) and a knowledge sorting process (Chen & Occeña, 1999) based on hierarchical sorts were employed for the analysis of knowledge attributes, such as the customer needs, standards, and ease of use.

Previous research work revealed that preference elicitation or the selection of design alternatives plays a crucial role in decision making during product conceptual design. Some of these attempts include utility analysis (Pahl & Beitz, 1984), decision networks (Garcia & Howard, 1992), fuzzy logic (Thurston & Carnahan, 1992), evidential reasoning (Yang & Sen, 1997), optimal design (Papalambros, 1995), flexible design (King & Sivaloganathan, 1999), graphical methods (Pugh, 1991), and eigenvector methods (Sen & Yang, 1995). Among them, Mitri (1991) employed a task-specific architecture for knowledge-based system development to evaluate different design options (or candidates) so as to select the best one of them. D'Ambrosio and Birmingham (1995) proposed a systematic methodology, known as preference-directed design, to generate an imprecise value function based on a set of preference statements that reflected the relative importance of each design attribute. Tseng and Du (1998) applied the conjoint analysis technique, which was coined by Green and Srinivasan (1978), to elicit the preference among the designing products using the mapping between functional requirements and the physical domain. Dentsoras (1999) postulated a selective multiple-criteria method for handling constraint violations to define product concepts derived from design parameter graphs.

The elicitation of customer requirements focuses on the conversion process that translates customer verbatim constructs, which are often tacit and subjective, into an explicit and objective customer requirement statement for use in the early stage of product concept generation. This paper proposes an integrated approach, which is based on repeated single-criterion sorts (Rugg et al., 1992) and fuzzy evaluation, to elicit and analyze customer requirements for NPD. In Section 2 the various sorting techniques for customer requirements acquisition are described. Section 3 presents the basic notions of fuzzy evaluation. The details of the integrated approach are described in Section 4. The approach is validated using a case study on the design of a wooden golf-club head. The details of the validation are presented in Section 5. The last section, Section 6, summarizes the main conclusions reached in this work.

2. SORTING TECHNIQUES FOR CUSTOMER REQUIREMENTS ACQUISITION

2.1. Background

Appropriate elicitation techniques that are able to offer a compromised solution between the extensiveness of expertise and the genuineness of the VoC are necessary for effective elicitation and evaluation of customer requirements. Many elicitation techniques were developed over the last century. These include ethnographic approaches (Mead, 1928), reports (Cortazzi & Roote, 1975), questionnaires (Krippendroff, 1980), and interviews (Gordon & Gill, 1991). Although the repertory grid technique (Shaw, 1980), which is derived from the attribute matrix, was extensively employed in knowledge acquisition, it is however unsuitable for dealing with those problems concerning nominal values, that is, nonscalar categories in relation to requirements elicitation. Basically, sorting techniques are derived from Kelly's personal construct theory (Kelly, 1955). The basic notions of sorting techniques involve a process where objects such as pictures and cards are sorted into groups by customers. As summarized by Rugg and McGeorge (1997), sorting techniques are multifunctional, flexible, easy to use, and systematic.

In general, sorting techniques can be classified into four broad categories, namely Q-sorts (Stephenson, 1953), hierarchical sorts (Rugg & McGeorge, 1995), "all in one" sorts (Benfer et al., 1991), and repeated single-criterion sorts (Rugg et al., 1992). In this work, the technique of repeated single-criterion sorts was adopted because it is more flexible and easier for most novice elicitors to handle. Using such a technique, the same pictures are repeatedly sorted each time when a different single attribute or a sorting criterion is used. More specifically, picture sorts that enable novice respondents to detect the slight difference between sorted pictures, and thereafter extract the cause for that effect, were chosen to elicit customer requirements.

2.2. Picture sorts for customer requirements elicitation

The picture sorts for customer requirements elicitation comprise five main steps: selecting the entities, preparing the pictures and instructions, conducting the session, recording the session, and analyzing the criteria and categories (Rugg & McGeorge, 1999). In picture sorts, the entities are the objects selected for formulating pictures, while the respondents (customers) can categorize these pictures into groups using a criterion or requirement each time for a sort. Table 1 shows the definitions of the terminology used in sorting techniques. These five main steps are now briefly described for clarity.

Step 1. *Selecting the entities*. The basic principle regarding the choice of entities involves two aspects, namely, appropriate semantic coverage and appropriate level of hier-

Table 1. *Definitions of terminology used in sorting techniques*

Terminology	Definition	Example
Construct	An attribute used by an individual to describe something	Light weight
Superordinate construct	The attribute grouped from the criteria in a higher abstraction level	Use new material to reduce weight
Imposed construct	The high-level attribute abstracted from the superordinate construct	Usability
Criterion	The attribute used as the basis for a sort when using a sorting technique, also called "verbatim construct"	Light weight is preferred
Category	A group into which things may be classified using a criterion, also called "subordinate construct"	Plastics, composite materials
Design alternative	A concept or solution in one subtree of a specific product	Light-weight type

archy. In the event of using picture sorts for customer requirements elicitation, it is more likely to select a set of new design alternatives belonging to one subtree of a specific product. The number of entities is recommended to be more than eight. For this purpose, laddering may be an adequate preprocessing technique for the choice of entities.

Step 2. *Preparing the pictures and instructions*. Pictures should conform to the same standards, that is, the same size, the same background, and similar glossiness. In addition, they need to be randomly numbered and free of extraneous features. An explanation of the picture sorting process for all the customers (respondents) to follow is desirable. During each sorting session, it is highly desirable to use the same set of instructions.

Step 3. *Conducting the session*. Once the pictures and instructions are available to the respondents, a sorting session is activated. During the sorting process, the respondents should sort pictures into groups with one group for each category and using only one criterion or requirement in each sort, which is repeated single-criterion sorts. In other words, a picture can only be sorted once into one of the categories elicited by respondents in each sort; meanwhile, the criterion can be treated as the grouping principle used by respondents. Upon completion, the respondents verbalize their constructs and name the categories. They must also ensure that there is no leftover, that is, no item not belonging to any of the groups. The dyadic (or triadic) elicitation technique is further used to elicit more constructs from explicit knowledge to semitacit or tacit knowledge of some kind, if necessary.

Step 4. *Recording the session*. To avoid mistakes in recording a session and to save recording time, it is advisable to use code numbers instead of meaningful names.

Step 5. *Analyzing the criteria/categories*. The analysis of the recording from a session generally depends on the purpose of the session. This includes the number of criteria, the type of criteria, the commonality of criteria, the distribution of commonality, and the analysis of categories. Basically, the number of criteria provides an indication about the range of knowledge within the population of respondents. There are three main types of criteria: observable or unobservable, objective or subjective, and intrinsic or extrinsic. If the criteria are unobservable, they may share different facets from the higher level of hierarchy. As for the commonality of criteria, it indicates the amount of overlap between respondents. Generally, high commonality indicates a smooth sorting process, whereas low commonality implies further work needed. The distribution of commonality indicates two situations: one involves much in the public domain, where the commonality distributes densely in some main criteria; and the other involves much in the individual domain, where the commonality distributes sparsely without any obvious pattern. In the latter situation the respondents are said to possess different tastes or different cultures. The analysis of categories is fairly similar to those for criteria. It includes the number of categories, the type of categories, the commonality of categories and the distribution of categories within a criterion.

Two sessions of a sorting process were implemented in this work. Prior to conducting a second session using pairwise comparisons, the statistical results obtained from the first session need to be treated. The treatment involves two steps:

- grouping the verbatim constructs into the superordinate constructs and
- defining the imposed constructs by domain experts.

This is to allow the distribution of commonality for the superordinate constructs and the imposed constructs to be studied. Subsequently, the pairwise comparison is used in the second session by the same groups of the respondents or customers. If the distribution of commonality between the groups is sparse, different patterns such as different tastes or color for a new product will then be considered. In this way, picture sorts provide a means to preprocess the complex VoC into useful information. Upon completion of the second session of the picture sorts, the superordinate constructs and the imposed constructs obtained are further processed by a two-level fuzzy evaluation process (FEP) described in Section 3.

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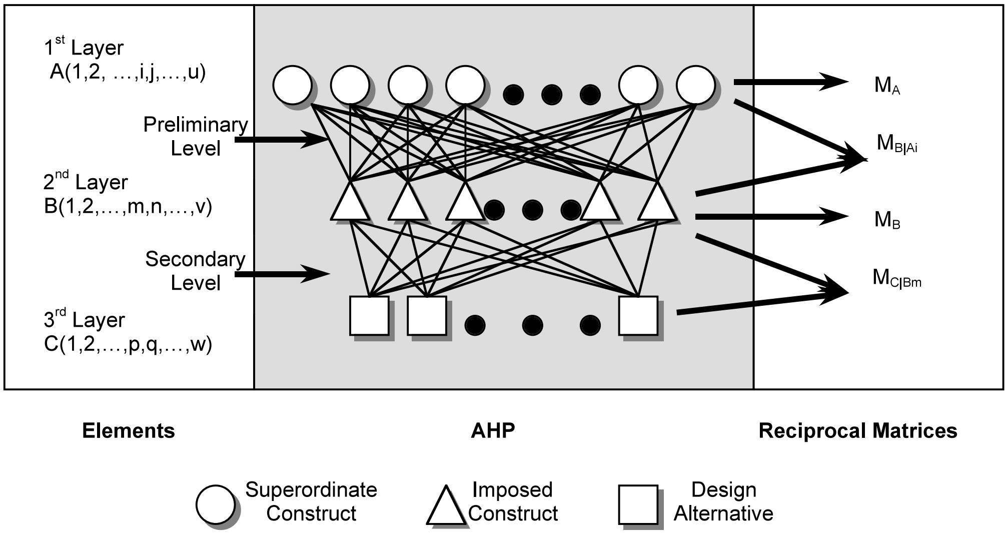

Fig. 1. The hierarchy of the objects in FEP.

3. QUANTITATIVE ANALYSIS OF CUSTOMER REQUIREMENTS USING FUZZY EVALUATION

Further investigations are needed to analyze the criteria or customer requirements, which are originally in the form of verbatim constructs. During these steps, it is necessary to extend and deepen the elicited customer requirements. However, when the pictures of design alternatives are being sorted, the majority of criteria elicited are expected to be nonscalar, as well as ambiguous and subjective in nature. Thus, it is problematic for picture sorts alone to systematically evaluate the elicited customer requirements because they are used for clear-cut decision making. Furthermore, subjective decision making, such as the preference of design alternatives, often contains ambiguity.

To overcome such a problem, the AHP proposed by Saaty (1988) was adapted for the evaluation of customer requirements and used as a fuzzy evaluation toolkit (Asai, 1995). This results in the establishment of a two-level FEP. Therein, several interrelated criteria like the superordinate constructs are considered simultaneously. Elements such as weights and evaluation values are indirectly elicited for the evaluation of qualitative items. In other words, the FEP provides a metric to handle inconsistency in human judgment. It involves the following steps.

Step 1. *Constructing a hierarchy of objects*. Figure 1 shows the FEP possesses a three-layer AHP structure. An imposed construct (B_m) is fully interconnected to all the superordinate constructs (A_i). This implies that the mapping between them is an AND relation. Accordingly, a design alternative (C_p) is also fully interconnected to all the imposed constructs defined by domain experts.

Step 2. *Applying pairwise comparisons*. In the first layer of the FEP, the pairwise comparison is first conducted between Element A_i and Element A_j of the superordinate constructs. The ratio of A_i for A_j (a_{ij}) which indicates which one is more important and by how much, is then obtained. If A_i is more important than A_j, then $a_{ij} > 1$. The opposite of a_{ij} (i.e., a_{ji}) is defined as

$$a_{ji} = 1/a_{ij}, \quad \text{for } i, j = 1, 2, \ldots, u, \tag{1}$$

where u is the number of elements within the layer of the superordinate constructs. Especially, $a_{ii} = 1$.

The importance that is expressed linguistically is converted to numerical values using the scheme (Table 2) proposed by Asai (1995). A $u \times u$ dimensional reciprocal matrix (M_A) is subsequently established [Eq. (2)] using all the pairwise comparisons between the superordinate constructs, rather than directly assigning weights or importance ratings for the superordinate constructs. As a result, in a reciprocal matrix it is much easier to detect the relative importance of the superordinate constructs than directly assigning weights for them.

$$M_A = (a_{ij}), \quad \text{for } i, j = 1, 2, \ldots, u. \tag{2}$$

Table 2. *List of conversions*

Numerical Value	Linguistic Expression
1	Equal importance
3	Weak importance
5	Strong importance
7	Demonstrated importance
9	Absolute importance

The numerical values 2, 4, 6, and 8 stand for intermediates. Adapted from Asai (1995).

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For the same reason, another $v \times v$ dimensional reciprocal matrix between imposed constructs (M_B) can be obtained for the second layer as follows:

$$M_B = (b_{mn}), \quad \text{for } m, n = 1, 2, \ldots, v, \tag{3}$$

where v is the number of elements in the layer of imposed constructs.

Next, pairwise comparisons are applied between the elements in the layer of the imposed constructs, such as B_m and B_n, with respect to each element in the layer of the superordinate constructs, such as A_i. As a result, reciprocal matrices, expressed as $M_{B|A_i}$ $(i = 1, 2, \ldots, u)$, can be acquired. The process determines the degrees of correlation between the superordinate constructs and the imposed constructs. This admits the inconsistency in judgments and provides a measure for inconsistency in judgments at the same time. By the same token, pairwise comparisons are also employed between elements in the layer of the design alternatives, such as C_p and C_q($p, q = 1, 2, \ldots, w$, where w is the number of elements in the layer of the design alternatives), with respect to each element in the layer of the imposed constructs, such as B_m. Therefore, reciprocal matrices, expressed as $M_{C|B_m}$ $(m = 1, 2, \ldots, v)$, can be obtained.

Step 3. *Proceeding to fuzzy evaluation*. After a reciprocal matrix such as M_B has been acquired, a v-dimensional priority eigenvector, φ_B, is deduced from Eqs. (4) and (5) by associating with a scalar, the maximum eigenvalue of that reciprocal matrix, $\gamma_{B_{\max}}$. Furthermore, it is obtained by normalizing the sum of elements to one, that is, $\sum_{m=1}^{v} \varphi_B(m) = 1$.

$$M_B \varphi_B = \gamma_{B_{\max}} \varphi_B, \tag{4}$$

$$\gamma_{B\max} \geq v \quad \text{iff } b_{mn} b_{no} = b_{mo}, \quad \text{for } m, n, o = 1, 2, \ldots, v, \tag{5}$$

where b_{mn} is the pairwise comparison ratio between the mth and nth imposed constructs, b_{no} is the pairwise comparison ratio between the nth and oth imposed constructs, and b_{mo} is the pairwise comparison ratio between the mth and oth imposed constructs.

To estimate the closeness between $\gamma_{B_{\max}}$ and v, which shows the degree of consistency for M_B, a consistency ratio (CR_B) is defined as follows:

$$\text{CR}_B = \frac{\gamma_{B_{\max}} - v}{(v-1)\text{RI}}, \tag{6}$$

where RI is the average random index adopted for all reciprocal matrices, and the recommended RI values are given in Table 3 (Asai, 1995).

If CR_B is around 0.1 or less, the reciprocal matrix M_B can be considered as consistent; otherwise, some adjustments to the subjective judgments are needed. Other priority eigenvectors, $\varphi_{B|A_i}$, $\varphi_{C|B_m}$, can be similarly obtained, along with their maximum eigenvalues, $\gamma_{B|A_{i\max}}$, $\gamma_{C|B_{m\max}}$, and consistency ratios, $\text{CR}_{B|A_i}$, $\text{CR}_{C|B_m}$. Accordingly, it implies that the larger the priority eigenvector is, the stronger the correlation between relevant elements can be.

Table 3. *List of average random index*

	Element Number				
	6	7	8	9	10
Average random index	1.24	1.32	1.41	1.45	1.49

The element number is the number of elements in each layer of the hierarchy. Adapted from Asai (1995).

Step 4. *Conducting decision making*. The FEP enables decision making to be performed in two levels. The preliminary level aims at identifying the degrees of correlation between the superordinate constructs and the imposed constructs, as well as between the imposed constructs and the design alternatives. This can be detected from the values of relevant priority eigenvectors, such as $\varphi_{B|A_i}$ and $\varphi_{C|B_m}$. Assume that all the elements in M_A are set to one. This implies that all the superordinate constructs are equally important.

At the secondary level of the FEP, the preference for the design alternatives can be obtained by examining the ranking of the product of eigenvectors, $\varphi'_{C|B_m} \varphi_B$. For example, the preferred design alternative #p, which is ranked first, is associated with the product of two different priority eigenvectors having the maximum value as follows:

$$\#p \rightarrow \text{first ranking}, \quad \text{if } \varphi'_{C|B_m} \varphi_B(p) \rightarrow \max. \tag{7}$$

4. AN INTEGRATED APPROACH TO THE SOLICITATION OF CUSTOMER REQUIREMENTS

An integrated approach that combines the strengths of the repeated single-criterion sorts with the FEP for the solicitation of customer requirements for engineering design is depicted in Figure 2. The approach comprises two integral phases, namely, a two-session picture sorts phase and a two-level fuzzy evaluation phase. Basically, sorting techniques, particularly picture sorts, depend on the way in which the sorting process is designed, conducted, and controlled by the elicitors. Undisputedly, the customers (respondents) play an important role in defining the initial groupings to customer requirements elicitation. It is apparent that the collaborative efforts of elicitors, respondents, and experts are needed during the two-session repeated single-criterion sorts phase.

During the first session, the elicitors present a series of pictures of a product to the respondents. Verbatim constructs or criteria, which contain the VoC, are then solicited from the respondents. With the assistance of domain experts, the verbatim constructs are first preprocessed into

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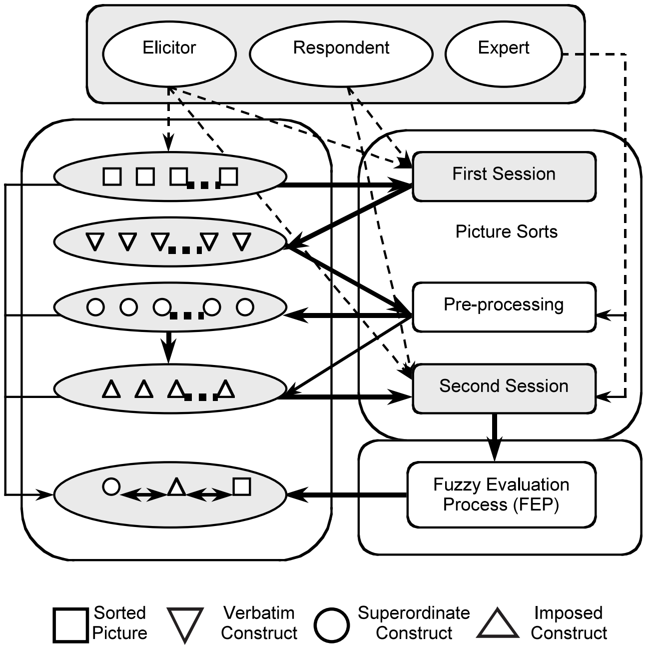

Fig. 2. An integrated approach.

superordinate constructs and subsequently into imposed constructs. Then the second session of picture sorts is initiated. Basically, the second session is complementarily conducted using pairwise comparisons for customer requirements. Upon completion of the picture sorts, the two-level fuzzy evaluation phase is activated to quantify the desired customer requirements.

5. A CASE STUDY

5.1. A golf-club head design

The case study involved nine randomly numbered pictures (#1–#9) of new design alternatives for a certain brand of golf clubs (Fig. 3). The pictures of these design alternatives were chosen from a series of computer-aided design (CAD) drawings, which were randomly numbered at their right-hand upper corner. Each picture was standardized and displayed in four views (isometric, top, front, and side).

Twelve respondents divided in three groups (Groups A–C) were asked to sort these pictures using repeated single-criterion sorts. Each group consisted of one female and three male subjects. In the first session, the respondents elicited their own criteria or categories (verbatim or subordinate constructs) using repeated single-criterion sorts. Subsequently, the superordinate constructs were preprocessed. The domain expert then proceeded to define a set of imposed constructs. During the second session, pairwise comparisons were conducted. Furthermore, a domain expert was also involved in the second session to ensure that important criteria of customer requirements were involved. Upon completion, the two-level fuzzy evaluation was invoked to quantify the customer requirements.

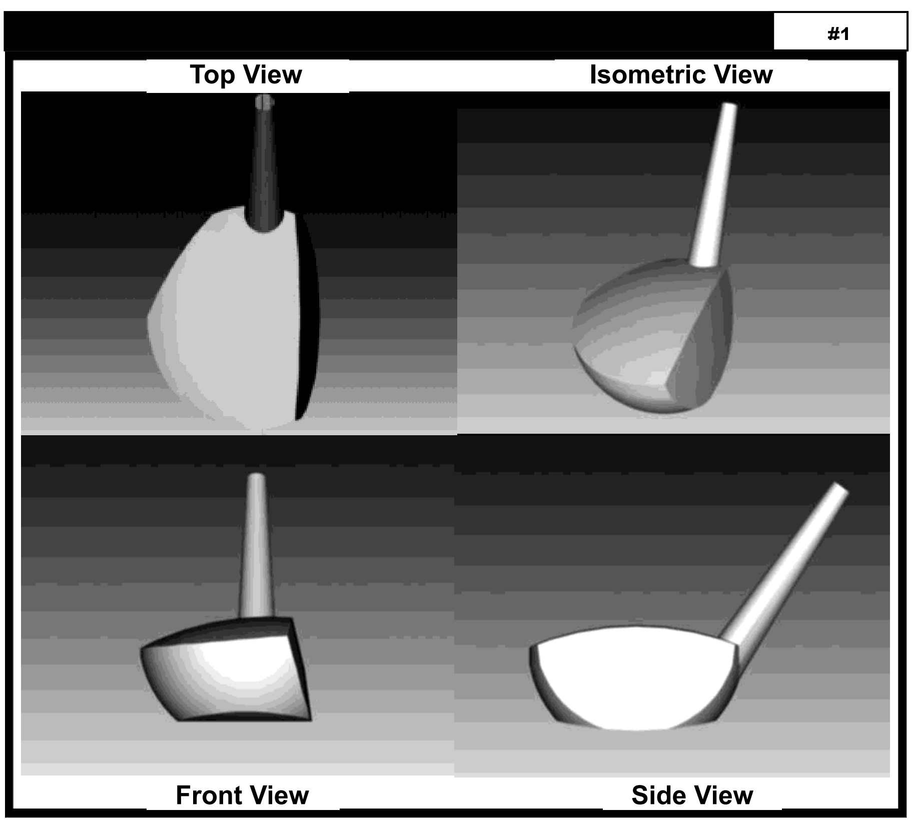

Fig. 3. A sample picture for picture sorts (four views of a CAD drawing).

Table 4. *Superordinate constructs generated by three groups of respondents*

Code	Superordinate Construct	Group of Respondents A	B	C
1	Likes the design	•		•
2	Provide more market information	•	•	•
3	Priced reasonably	•	•	
4	Cost as low as possible	•	•	•
5	Follows design standards	•		•
6	Use standard parts	•		
7	Provide fashionable styles	•		•
8	Possess good appearance			•
9	Use new materials to reduce weight, for example		•	
10	Can be used for a long time	•	•	
11	Possess desirable mechanism (flex, swing weight)	•		
12	Is easy to manufacture	•		•
13	Is easy of assemble/repair	•		
14	Is easy to use	•	•	•
15	Provide adequate dimensions (e.g., the length)	•	•	•
16	Design to suit specific customer	•		•
Total	16 superordinate constructs	14	7	10

(•) The elicitation result from each group of respondents.

Table 5. *List of imposed constructs*

Code	Imposed Construct
I	Market or business
II	Cost or price
III	Design principles
IV	Manufacture or assemble or repair
V	Durability or usability
VI	Personal preferences

Table 4 lists the superordinate constructs and their commonality distribution between each group of respondents. We found that a high commonality of superordinate constructs existed that was elicited from the three groups of subjects. This was probably due to the fact that most of the subjects from Group A were familiar with product design and manufacturing. It also appeared that Group B scored more on aesthetic attributes and Group C attained high scores for materials-related attributes. A domain expert then grouped these superordinate constructs into a set of imposed constructs (Table 5) for investigation in the second session.

During the two-level fuzzy evaluation, the correlation between the superordinate constructs and the imposed constructs, which is the categorization of the superordinate constructs with regard to the imposed constructs, was investigated. All the superordinate constructs were assumed to have equal importance because of the trade-off between the high commonality distribution and expert judgment. The reciprocal matrices $M_{B|A_i}$ (Table 6) were then obtained by performing pairwise comparisons between the imposed constructs (6 of them, Codes I–VI in Table 6) in relation to the superordinate constructs (16 of them). These matrices were initially elicited by the respondents and finalized by the domain expert so that some kind of trade-off on the results

5.2. Results and discussion

There were 39 verbatim constructs (criteria) together with their respective subordinate constructs (categories) elicited in the first session of picture sorts. We observed that a large number of the constructs that were elicited were subjective, unobservable, or abstract ones because some of the respondents had prior knowledge about the specific brand of golf club and its manufacturing process. Superordinate constructs were extracted to identify the commonality distribution and to avoid intensive processing.

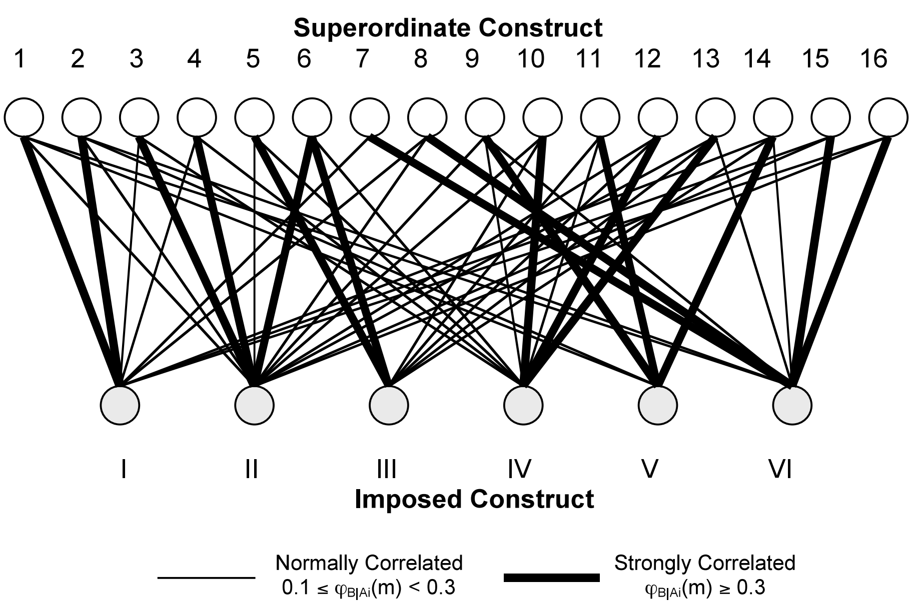

Fig. 4. The correlation between superordinate and imposed constructs.

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Table 6. *Reciprocal matrices between imposed constructs*

	I	II	III	IV	V	VI	I	II	III	IV	V	VI	I	II	III	IV	V	VI
$M_{B\|A_i\|}$	Superordinate Construct 1						Superordinate Construct 2						Superordinate Construct 3					
I	1	3	7	9	3	1	1	5	7	9	4	2	1	1/3	5	1	3	7
II	1/3	1	2	5	1	1/3	1/5	1	5	7	1/2	1/3	3	1	7	5	7	9
III	1/7	1/2	1	3	1/2	1/4	1/7	1/5	1	2	1/3	1/7	1/5	1/7	1	1/5	1/3	1/2
IV	1/9	1/5	1/3	1	1/5	1/9	1/9	1/7	1/2	1	1/5	1/9	1	1/5	5	1	5	7
V	1/3	1	2	5	1	1/3	1/4	2	3	5	1	1/3	1/3	1/7	3	1/5	1	5
VI	1	3	4	9	3	1	1/2	3	7	9	3	1	1/7	1/9	2	1/7	1/5	1
	Superordinate Construct 4						Superordinate Construct 5						Superordinate Construct 6					
I	1	1/5	1	2	5	7	1	1/3	1/7	1/5	1	1/2	1	1/7	1/7	1/5	1/2	3
II	5	1	7	3	7	9	3	1	1/3	1/2	5	7	7	1	1	2	7	9
III	1	1/7	1	1/3	1	5	7	3	1	2	7	9	7	1	1	2	7	9
IV	1/2	1/3	3	1	5	7	5	2	1/2	1	5	7	5	1/2	1/2	1	5	7
V	1/5	1/7	1	1/5	1	3	1	1/5	1/7	1/5	1	3	2	1/7	1/7	1/5	1	3
VI	1/7	1/9	1/5	1/7	1/3	1	2	1/7	1/9	1/7	1/3	1	1/3	1/9	1/9	1/7	1/3	1
	Superordinate Construct 7						Superordinate Construct 8						Superordinate Construct 9					
I	1	5	7	7	7	1/3	1	3	5	7	5	1/5	1	1/3	3	1/5	1/7	1/5
II	1/5	1	2	5	1	1/7	1/3	1	5	7	5	1/5	3	1	5	1	1/2	1
III	1/7	1/2	1	2	1	1/7	1/5	1/5	1	1/3	1	1/7	1/3	1/5	1	1/5	1/7	1/5
IV	1/7	1/5	1/2	1	1	1/9	1/7	1/7	3	1	1/2	1/9	5	1	5	1	1/2	1
V	1/7	1	1	1	1	1/7	1/5	1/5	1	2	1	1/7	7	2	7	2	1	2
VI	3	7	7	9	7	1	5	5	7	9	7	1	5	1	5	1	1/2	1
	Superordinate Construct 10						Superordinate Construct 11						Superordinate Construct 12					
I	1	1/2	1/3	1/7	1	5	1	1/3	1/7	1/5	1/7	1/9	1	1/5	1/5	1/7	2	3
II	2	1	2	1/3	5	7	3	1	1/5	1/5	1/7	1/2	5	1	1	1/2	5	7
III	3	1/2	1	1/5	2	5	7	5	1	3	1/2	7	5	1	1	1/3	5	7
IV	7	3	5	1	7	9	5	5	1/3	1	1/3	5	7	2	3	1	7	9
V	1	1/5	1/2	1/7	1	2	7	7	2	3	1	9	1/2	1/5	1/5	1/7	1	2
VI	1/5	1/7	1/5	1/9	1/2	1	9	2	1/7	1/5	1/9	1	1/3	1/7	1/7	1/9	1/2	1
	Superordinate Construct 13						Superordinate Construct 14						Superordinate Construct 15					
I	1	1/3	1/5	1/7	3	1/2	1	5	2	5	1/5	1	1	3	7	8	9	1/2
II	3	1	2	1/3	7	3	1/5	1	1/3	2	1/7	1/5	1/3	1	5	7	7	1/5
III	5	1/2	1	1/2	5	2	1/2	3	1	5	1/3	1/2	1/7	1/5	1	3	2	1/7
IV	7	3	2	1	9	4	1/5	1/2	1/5	1	1/9	1/7	1/8	1/7	1/3	1	1/2	1/9
V	1/3	1/7	1/5	1/9	1	1/5	5	7	3	9	1	1/2	1/9	1/7	1/2	2	1	1/7
VI	1	1/3	1/2	1/4	5	1	1	5	2	7	2	1	2	5	7	9	7	1
	Superordinate Construct 16																	
I	1	3	6	7	9	1/3												
II	1/3	1	5	7	9	1/7												
III	1/6	1/5	1	3	3	1/7												
IV	1/7	1/7	1/3	1	1/3	1/9												
V	1/9	1/9	1/3	3	1	1/5												
VI	3	7	7	9	5	1												

The numerals I–VI denote the code number of the imposed constructs; $M_{B|A_i}$, the reciprocal matrices between imposed constructs in relation to each superordinate construct.

could be made. Consequently, the priority eigenvectors $\varphi_{B|A_i}$, the maximum eigenvalues $\gamma_{B|A_{i\max}}$, and the consistency ratios $\mathrm{CR}_{B|A_i}$ were obtained.

Figure 4 and Table 7 show the following information:

- Each superordinate construct may have different degrees of correlation with several imposed constructs. For example, the superordinate construct, "Likes the design," has a strong relation ($\varphi_{B|A_i}(m) \geq 0.3$) with such imposed constructs as "Market/Business" and "Personal Preferences," which have the values of 0.3499 and 0.3142 from the priority eigenvector, respectively. It possesses a normal relation ($0.1 \leq \varphi_{B|A_i}(m) < 0.3$) with imposed constructs "Cost/Price" and "Durability/Usability" because both values from the priority eigenvector are 0.1213.

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Table 7. *Results from preliminary level of fuzzy evaluation*

$\varphi_{B\|A_i\|}$	I	II	III	IV	V	VI	$\gamma_{B\|A_{i\max}}$	$\mathrm{CR}_{B\|A_i}$
1	0.3499[a]	0.1213[b]	0.0654	0.0279	0.1213[b]	0.3142[a]	6.3132	0.0505
2	0.4128[a]	0.1157[b]	0.0407	0.0263	0.1283[b]	0.2763[b]	6.2091	0.0337
3	0.1799[b]	0.4820[a]	0.0340	0.1939[b]	0.0770	0.0337	6.3919	0.0632
4	0.1770[b]	0.4815[a]	0.0879	0.1731[b]	0.0543	0.0253	6.4466	0.0720
5	0.0472	0.1840[b]	0.4091[a]	0.2599[b]	0.0586	0.0412	6.3101	0.0500
6	0.0469	0.3325[a]	0.3325[a]	0.2022[b]	0.0591	0.0267	6.1214	0.0196
7	0.2945[b]	0.0856	0.0518	0.0359	0.0532	0.4790[a]	6.2617	0.0422
8	0.2212[b]	0.1501[b]	0.0363	0.0410	0.0459	0.5055[a]	6.5873	0.0947
9	0.0548	0.1738[b]	0.0338	0.1967[b]	0.3443[a]	0.1967[b]	6.4097	0.0661
10	0.0802	0.2091[b]	0.1345[b]	0.4861[a]	0.0603	0.0298	6.2245	0.0362
11	0.0275	0.0452	0.2898[b]	0.1650[b]	0.3999[a]	0.0725	6.4405	0.0710
12	0.0626	0.2294[b]	0.2203[b]	0.4145[a]	0.0446	0.0286	6.5865	0.0946
13	0.0582	0.2290[b]	0.1870[b]	0.4066[a]	0.0287	0.0904[b]	6.5091	0.0821
14	0.1731[b]	0.0437	0.1143[b]	0.0293	0.3554[a]	0.2842[b]	6.4135	0.0667
15	0.2975[b]	0.1608[b]	0.0523	0.0266	0.0360	0.4268[a]	6.2344	0.0378
16	0.2531[b]	0.1548[b]	0.0539	0.0246	0.0384	0.4752[a]	6.6148	0.0992
Sum	2.7364	3.1985	2.1436	2.7096	1.9503	3.3061		

The numbers 1–16 show the code number of the superordinate constructs defined in accordance with Table 3; numerals I–VI are the code numbers of imposed constructs defined in accordance with Table 4.

$\varphi_{B|A_i}$, the priority eigenvectors between the superordinate constructs and the imposed constructs of reciprocal matrices $M_{B|A_i}$; $\gamma_{B|A_{i\max}}$, the maximum eigenvalues of $M_{B|A_i}$; $CR_{B|A_i}$ are the consistency ratios of $M_{B|A_i}$.

[a]A strong correlation between superordinate constructs and imposed constructs.

[b]A normal correlation between superordinate constructs and imposed constructs.

- The priority of the superordinate constructs in relation to each imposed construct can also be deduced. For example, if "Design Principles" is desired, it is advisable to emphasize "Follows design standards" and "Use standard parts" because the values from both priority eigenvectors are 0.4091 and 0.3325, respectively.
- The overall preference of the imposed constructs is obtained by summing up the values from all the priority eigenvectors with respect to each imposed construct. For example, "Personal Preferences" and "Cost/ Price" are ranked the top two among the imposed constructs. The sums of the values from those priority eigenvectors are 3.3061 and 3.1985, respectively.

The overlap of the superordinate constructs (Fig. 4) in relation to the imposed constructs reveals that the commonality distribution of the superordinate constructs exists. Figure 5 shows the 3-dimensional (3-D) space of the priority eigenvectors $\varphi_{B|A_i}$ and the top view of its contour diagram. It is observed that the peaks of the eigenvector space, which represent the strong correlation between the superordinate constructs and the imposed constructs, distribute mainly in the lower left-hand portion and the upper right-hand portion of the 3-D contour diagram. Further investigation shows that the subjects of all the groups are more concerned with personal and economic factors than issues such as durability.

In the same manner, the reciprocal matrices $M_{C|B_m}$, for the pairwise comparisons of the design alternatives in relation to the imposed constructs were obtained (Table 8). In addition, the reciprocal matrix M_B, which shows that an unequal importance among the imposed constructs exists according to the statistical results and expert judgment,

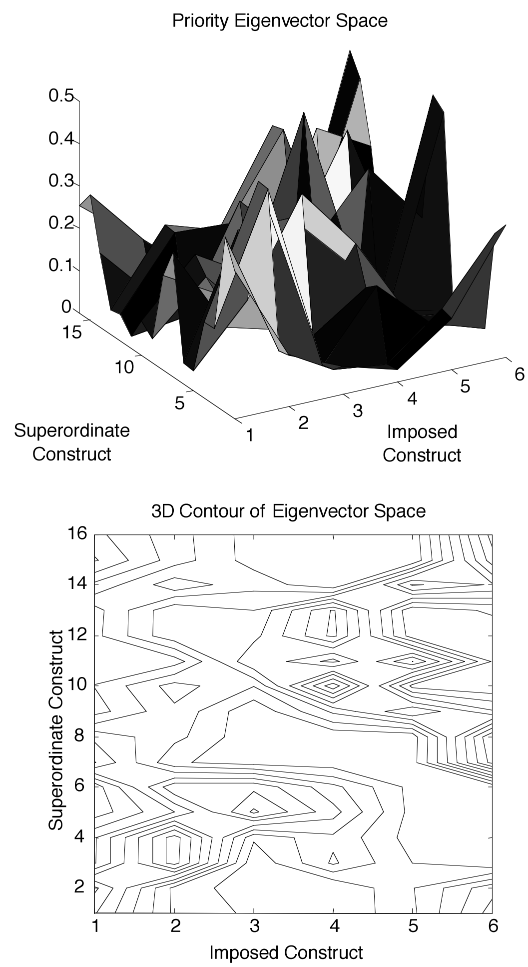

Fig. 5. A space diagram of $\varphi_{B|A_i}$.

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Table 8. *Reciprocal matrices between design alternatives*

$M_{C\|B_m}$	#1	#2	#3	#4	#5	#6	#7	#8	#9	#1	#2	#3	#4	#5	#6	#7	#8	#9
	Imposed Construct I									Imposed Construct II								
#1	1	4	2	2	5	2	4	8	8	1	8	7	2	7	1	6	7	9
#2	1/4	1	1	1/2	1	2	1	3	3	1/8	1	1/3	1/5	1/2	1/7	1/4	1/3	1/2
#3	1/2	1	1	1	1	1/2	2	6	6	1/7	3	1	1/3	1	1/7	1/4	1/2	1
#4	1/2	2	1	1	2	5	3	5	5	1/2	5	3	1	2	1/2	2	6	6
#5	1/5	1	1	1/2	1	3	2	4	4	1/7	2	1	1/3	1	1/4	1	1/3	1/2
#6	1/2	2	2	1/5	1/3	1	2	7	7	1	7	7	2	4	1	5	7	8
#7	1/4	1	1/2	1/3	1/2	1/2	1	9	9	1/6	4	4	1/2	1	1/5	1	2	1
#8	1/8	1/3	1/6	1/5	1/4	1/7	1/9	1	1	1/7	3	2	1/6	3	1/7	1/2	1	1/2
#9	1/8	1/3	1/6	1/5	1/4	1/7	1/9	1	1	1/9	2	1	1/6	2	1/8	1	2	1
	Imposed Construct III									Imposed Construct IV								
#1	1	2	3	1	3	1/2	5	7	7	1	5	5	1	3	1/3	3	1	2
#2	1/2	1	1	1/2	1	1/5	1/3	3	3	1/5	1	1/2	1/5	1/2	1/7	1/3	1/5	1/4
#3	1/3	1	1	1/3	1	1/6	1/2	6	6	1/5	2	1	1/5	1	1/7	1/4	1/5	1/5
#4	1	2	3	1	5	1	2	8	8	1	5	5	1	3	1/2	2	2	2
#5	1/3	1	1	1/5	1	1/5	1/2	4	4	1/3	2	1	1/3	1	1/4	1/2	1/3	1/4
#6	2	5	6	1	5	1	3	5	5	3	7	7	3	4	1	8	4	6
#7	1/5	3	2	1/2	2	1/3	1	2	2	1/3	3	4	1/2	2	1/8	1	1/4	1/3
#8	1/7	1/3	1/6	1/6	1/4	1/4	1/5	1	1	1	5	5	1/2	3	1/4	4	1	1/2
#9	1/7	1/3	1/6	1/8	1/4	1/5	1/2	1	1	1/2	4	5	1/2	4	1/6	3	2	1
	Imposed Construct V									Imposed Construct VI								
#1	1	1/5	1/3	1/2	1/7	1	1/3	1	1	1	1/5	1/7	1	1/6	1/2	1/4	1/9	1/8
#2	5	1	2	5	1	3	2	2	3	5	1	1/2	3	1/2	5	2	1/3	1/3
#3	3	1/2	1	3	1	5	4	3	3	7	2	1	5	1	7	3	1/2	1/2
#4	2	1/5	1/3	1	1/3	1	2	2	2	1	1/3	1/5	1	1/3	1	1/2	1/8	1/7
#5	7	1	1	3	1	5	6	3	4	6	2	1	3	1	7	5	1	1
#6	1	1/3	1/5	1	1/5	1	2	1	1	2	1/5	1/7	1	1/7	1	1/2	1/6	1/7
#7	3	1/2	1/5	1	1/5	1	2	1	1	4	1/2	1/3	2	1/5	2	1	1/3	1/3
#8	1	1/2	1/3	1/2	1/3	1	2	1	1	9	3	2	8	1	6	3	1	1
#9	1	1/3	1/3	1/2	1/4	1	2	1	1	8	3	2	7	1	7	3	1	1

The numbers 1–9 are the code numbers of design alternatives; $M_{C|B_m}$, the reciprocal matrices between design alternatives in relation to each imposed construct.

was also elicited (Table 9). The consequence of decision making on the preference of the design alternatives was then carried out by associating them with the priority eigenvectors φ_B and $\varphi_{C|B_m}$ (Table 10). Table 10 shows that Design Alternative #6 was the top choice according to the eigenvector product $\varphi_{C|B_m}\varphi_B$, which was computed as follows:

$$\#6 = 0.1143 \times 0.1356 + 0.2699 \times 0.2591 + 0.2575 \times 0.1356 + 0.3453 \times 0.1356 + 0.0603 \times 0.0749 + 0.0284 \times 0.2591 = 0.1790 \text{ (ranked first).} \tag{7}$$

Table 9. *Reciprocal matrix between imposed constructs*

M_B	I	II	III	IV	V	VI
I	1	1/2	1	1	2	1/2
II	2	1	2	2	3	1
III	1	1/2	1	1	2	1/2
IV	1	1/2	1	1	2	1/2
V	1/2	1/3	1/2	1/2	1	1/3
VI	2	1	2	2	3	1

Numerals I–VI are the code numbers of imposed constructs; M_B, the reciprocal matrix between imposed constructs.

Figure 6 shows the correlation between the imposed constructs and the design alternatives. The different levels of correlation between them, namely, weakly correlated ($\varphi_{C|B_m}(p) < 0.1$), moderately correlated ($0.1 \leq \varphi_{C|B_m}(p) < 0.2$), and strongly correlated ($\varphi_{C|B_m}(p) \geq 0.2$), can also be observed. This is the illustration of the priority eigenvectors, $\varphi_{C|B_m}$. Figure 7 illustrates the 3-D space of the priority eigenvectors $\varphi_{C|B_m}$ and the top view of its contour diagram. The peaks of the eigenvector space, which represent the strong correlation between the imposed constructs and the design alternatives, concentrate in the lower left-hand portion, lower right-hand portion, and the middle por-

Table 10. *Results from the secondary level of fuzzy evaluation*

$\varphi_{C\|B_m}$	I	II	III	IV	V	VI	$\varphi'_{C\|B_m}\varphi_B$
#1	0.2656[a]	0.2978[a]	0.1966[b]	0.1338[b]	0.0419[c]	0.0215[c]	0.1667 (2nd)
#2	0.0903[c]	0.0233[c]	0.0656[c]	0.0243[c]	0.2846[a]	0.0945[c]	0.0763 (8th)
#3	0.1080[b]	0.0390[c]	0.0746[c]	0.0290[c]	0.1850[b]	0.1549[b]	0.0928 (7th)
#4	0.1859[b]	0.1562[b]	0.1985[b]	0.1458[b]	0.0794[c]	0.0310[c]	0.1263 (3rd)
#5	0.1126[b]	0.0418[c]	0.0627[c]	0.0404[c]	0.2324[a]	0.1822[b]	0.1047 (4th)
#6	0.1143[b]	0.2699[a]	0.2575[a]	0.3453[a]	0.0603[c]	0.0284[c]	0.1790 (1st)
#7	0.0812[c]	0.0716[c]	0.0929[c]	0.0564[c]	0.0719[c]	0.0582[c]	0.0703 (9th)
#8	0.0211[c]	0.0504[c]	0.0255[c]	0.1106[b]	0.0648[c]	0.2155[a]	0.0951 (5th)
#9	0.0211[c]	0.0500[c]	0.0263[c]	0.1143[b]	0.0597[c]	0.2137[a]	0.0947 (6th)
$\gamma_{C\|B_{m\max}}$	10.0579	9.5519	9.5402	9.5461	10.1288	9.5026	
$\mathrm{CR}_{C\|B_m}$	0.0912	0.0476	0.0466	0.0471	0.0973	0.0433	
φ_B	0.1356 (3rd)	0.2591 (1st)	0.1356 (3rd)	0.1356 (3rd)	0.0749 (6th)	0.2591 (1st)	
$\varphi_{B_{\max}}$	6.0347						
CR_B	0.0056						

The numbers 1–9 are the code numbers of design alternatives; numerals I–VI are the code numbers of the imposed constructs; $\varphi_{C|B_m}$ are the priority eigenvectors between imposed constructs and design alternatives of the reciprocal matrices $M_{C|B_m}$; $\varphi_{C|B_{m\max}}$, the maximum eigenvalues of $M_{C|B_m}$; $\mathrm{CR}_{C|B_m}$, the consistency ratios of $M_{C|B_m}$; φ_B, the priority eigenvector between imposed constructs of the reciprocal matrix M_B; $\gamma_{B_{\max}}$, the maximum eigenvalue of M_B; CR_B, the consistency ratio of M_B.

[a]A strong correlation between imposed constructs and design alternatives.

[b]A moderate correlation between imposed constructs and design alternatives.

[c]A weak correlation between imposed constructs and design alternatives.

tion of the 3-D contour diagram. We observed that Imposed Constructs "Cost/Price" and "Personal Preferences" have the most influence in this study, which also agrees with the importance level of imposed constructs that can be deduced from the priority eigenvector φ_B in Table 10. Hence, the overall ranking of design alternatives is dependent on both priority eigenvectors of φ_B and $\varphi_{C|B_m}$, that is, the association of both eigenvectors. Accordingly, it can be found that Design Alternative #6 correlates strongly with more imposed constructs than any other design alternative does.

From this case study it is apparent that customer requirements elicitation plays an important role in the product design process. Basically, the involvement of customers in a design process includes the following: (1) during picture sorts, customers (respondents) contribute their criteria or requirements for establishing a three-layer hierarchy, which is used for further fuzzy evaluation; and (2) in the process of a two-level fuzzy evaluation, customers contribute their knowledge in the formation of the reciprocal matrices, which dominate the results from the fuzzy evaluation.

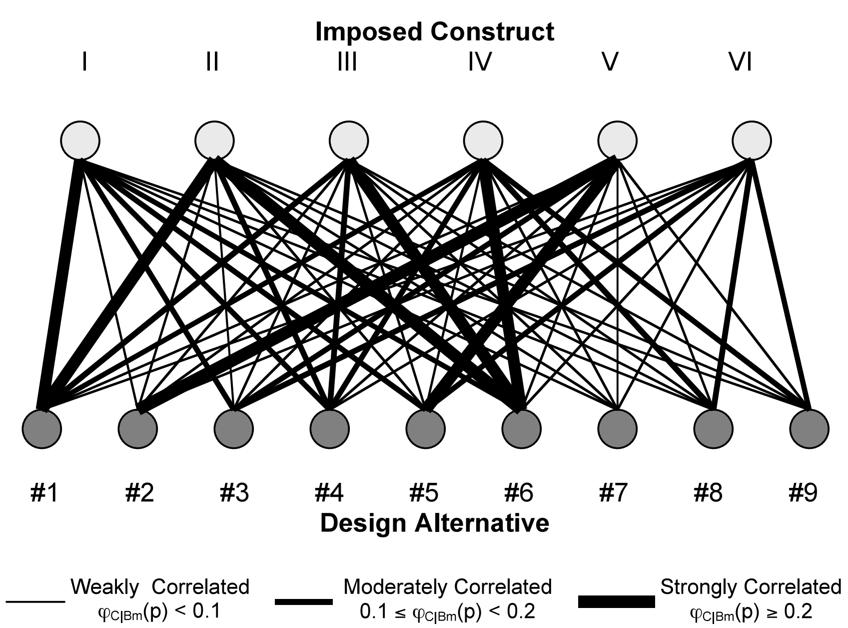

Fig. 6. The correlation between imposed constructs and design alternatives.

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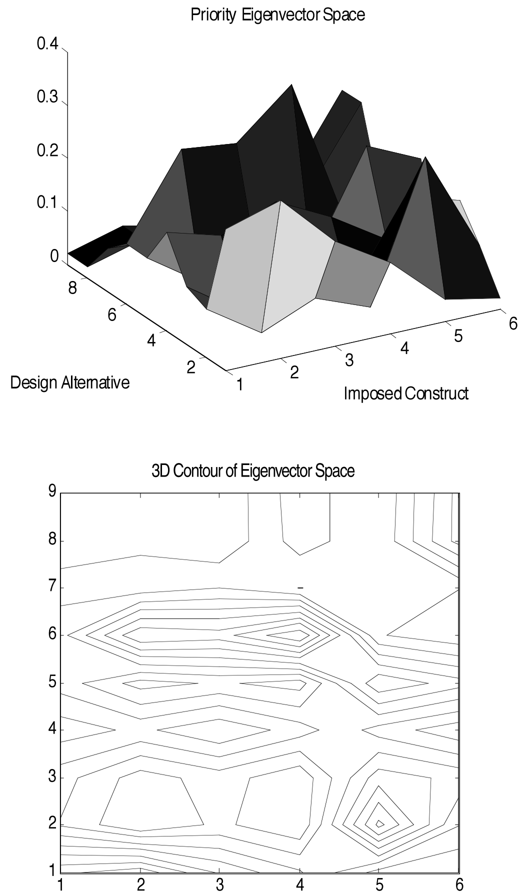

Fig. 7. A space diagram of $\varphi_{C|B_m}$.

As such, it can be inferred that (1) customer verbatim constructs, which are usually tacit and subjective, can be translated into an explicit and objective customer requirements statement for use via the quantitative fuzzy evaluation; and (2) the preferred design alternative elicited is close to market orientations and customer demands due to the intensive customer involvement in this approach.

6. CONCLUSIONS

An integrated approach based on repeated single-criterion sorts and fuzzy evaluation was proposed and implemented. The feasibility of using picture sorts, usually employed for knowledge acquisition, for customer requirements elicitation was demonstrated. The advantages of applying a sorting technique may include (1) broad coverage of domain for requirements elicitation, (2) time savings for both preparation and acquisition sessions, (3) less expert guidance during requirements elicitation, and (4) a standardized format that is fit for computerized automation and (5) friendly to elicitors and respondents (Maiden & Rugg, 1996). To overcome the qualitative and imprecise nature of the constructs elicited, a two-level FEP, which is based on the AHP, was developed to evaluate these constructs quantitatively. This two-level FEP enables different design alternatives to be prioritized and results in resolving ambiguity and subjective inherence of customer requirements during further quantitative evaluation. It also allows both exploratory and in-depth perspectives of customer requirements to be elicited and managed.

A case study on a wooden golf-club head design was used to illustrate the integrated approach we developed. Although a high commonality distribution of constructs was observed in this case study, this could be attributable to most of the respondents possessing prior knowledge about the subject matter. Nevertheless, picture sorts appear to be a feasible technique for the elicitation of customer requirements. Future work may include validation of the integrated approach using a larger sample size of respondents for other case studies and investigations into the possibility of automating the role of domain experts using artificial intelligence techniques such as knowledge-based systems. In this respect, the work by Chen et al. (2001*b*) can possibly be adapted. In conclusion, with the implementation of such an integrated approach to customer requirements elicitation, it is envisaged that genuine VoC can be incorporated into the design of products.

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